

Olfaction

Do.
Make.
Heal.

Innovate.

Reinvent the Future.



- Parkinson patients show olfactory deficits 4 years before showing motor symptoms.



Source: http://media.neurologyadvisor.com/images/2015/11/17/womanflowers_869683.jpg?format=jpg&zoom=1&quality=70&anchor=middlecenter&width=320&mode=pad

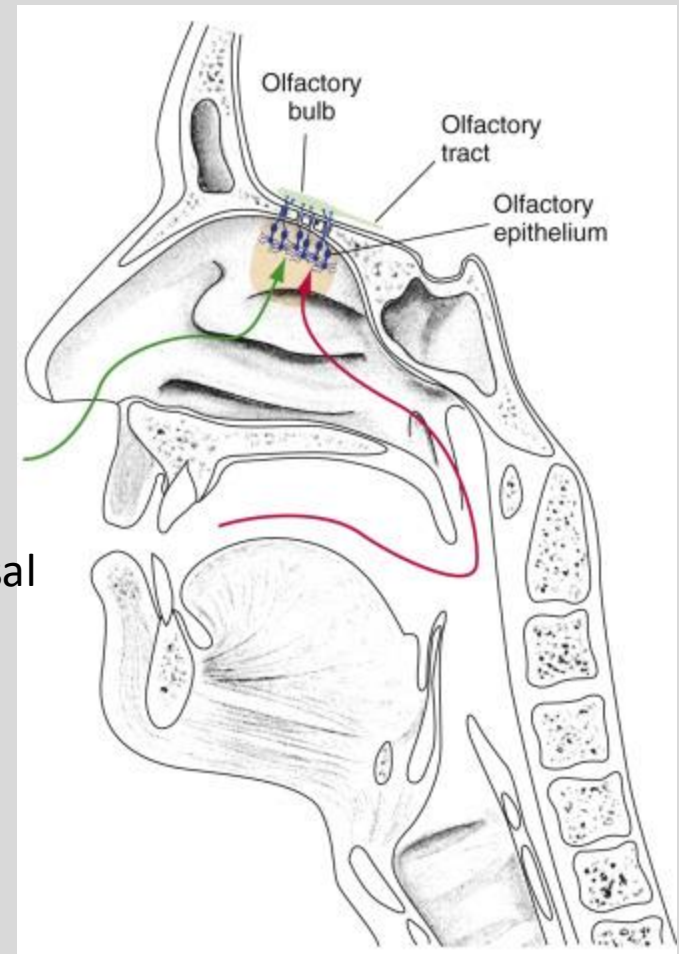


The Human Olfactory System

- *Retronasal olfaction* is the perception of odors emanating from the oral cavity during eating and drinking, as opposed to *orthonasal olfaction*, which occurs during sniffing.
- Retronasal olfaction is important for the flavor sensation= taste+olfaction

orthonasal

retronasal

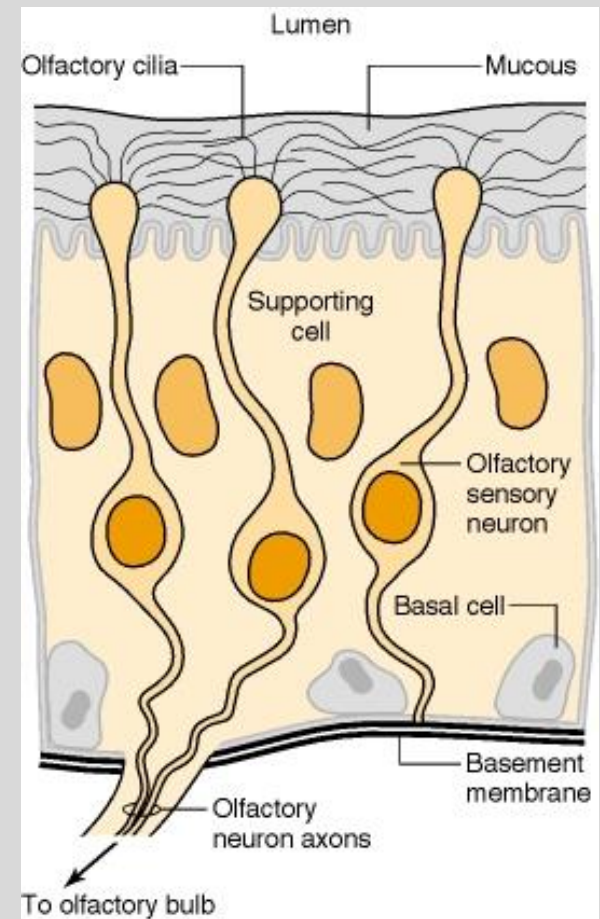


Olfactory Cells

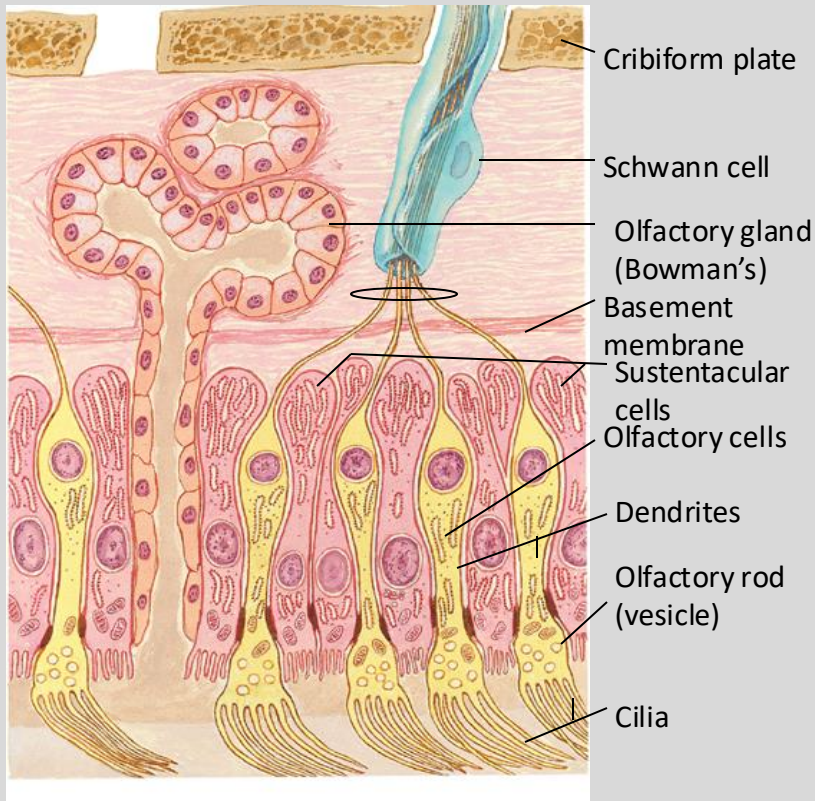
- First order bipolar neurons
- A.K.A. Olfactory neurosensory cells, primary olfactory neurons
- Dendrites with specialized cilia
Short-lived, high turnover (7 days)
- Chemotherapy affects smell perception

Basal cells:

- Stem cells that give rise to the olfactory receptor cells



Olfactory Epithelium - Supporting Cells

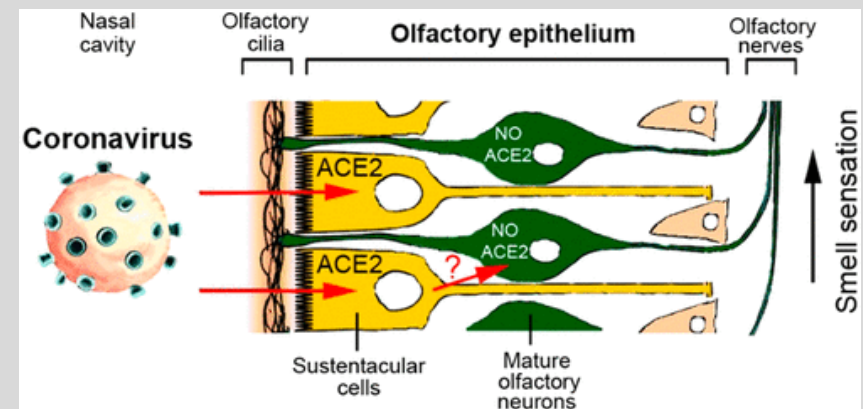


Sustentacular cells:

- Microvilli and secretory granules
- Mucous
- Structural stability

Cells associated with Bowman's glands:

- Secrete mucous fluid containing glycoproteins

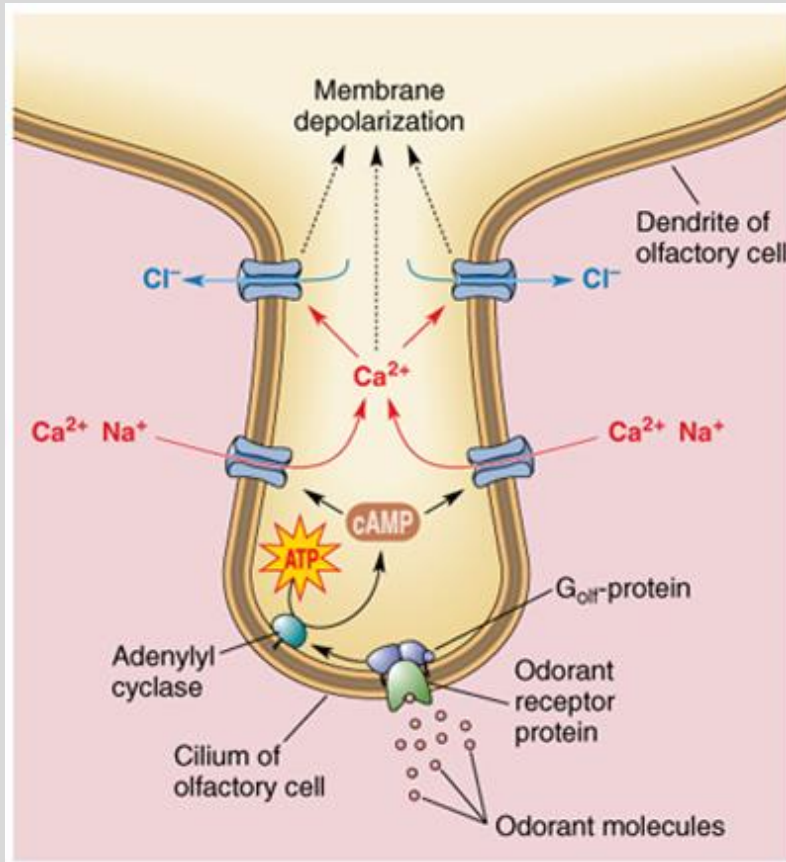


Expression of the SARS-CoV-2 Entry Proteins, ACE2 and TMPRSS2, in Cells of the Olfactory Epithelium: Identification of Cell Types and Trends with Age

Katarzyna Bilinska ,Patrycja Jakubowska ,Christopher S. Von Bartheld , and Rafal Butowt*
ACS Chem. Neurosci. 2020,



- Olfactory receptors are G-coupled protein receptors



- Olfactory receptor protein binds an array of odorants

- Humans have ~300 olfactory receptor genes (compared to mice ~1200)

- .-A receptor-odorant complex is formed, which activates a G protein.

- The second messenger cascade causes the opening of cation channels (Na⁺, K⁺ and Ca²⁺)

- Ca²⁺ influx causes the opening of Ca²⁺-gated Cl⁻ channels opening Cl⁻ channels causes depolarization of an olfactory receptor neuron and amplifies the receptor potential. The depolarization spreads to the trigger zone of the receptor neuron and, if large enough, initiates action potentials

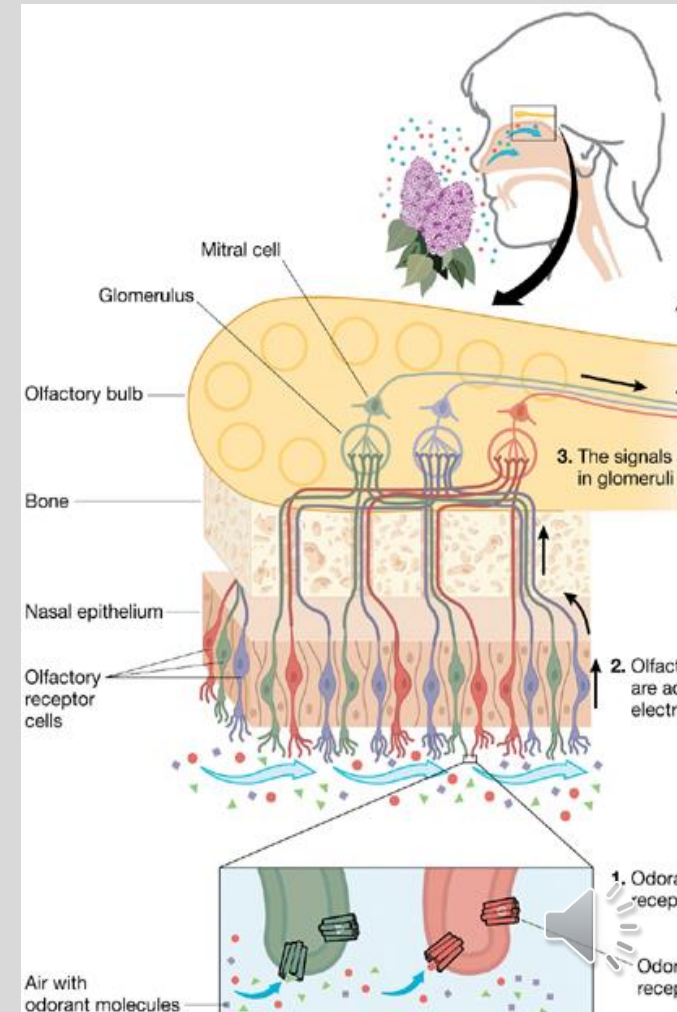


Olfactory Nerve: CNI

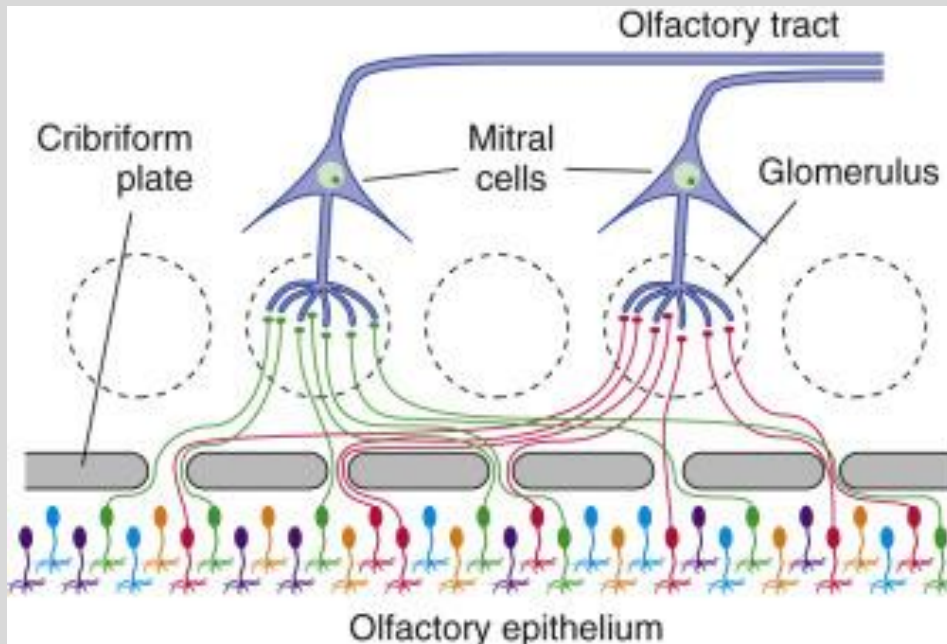
Unmyelinated axons of the olfactory cells gather into about 20 bundles - the olfactory fila

Traverse through foramina in the cribriform plate in the ethmoid bone

Synapse in the olfactory bulb in glomeruli



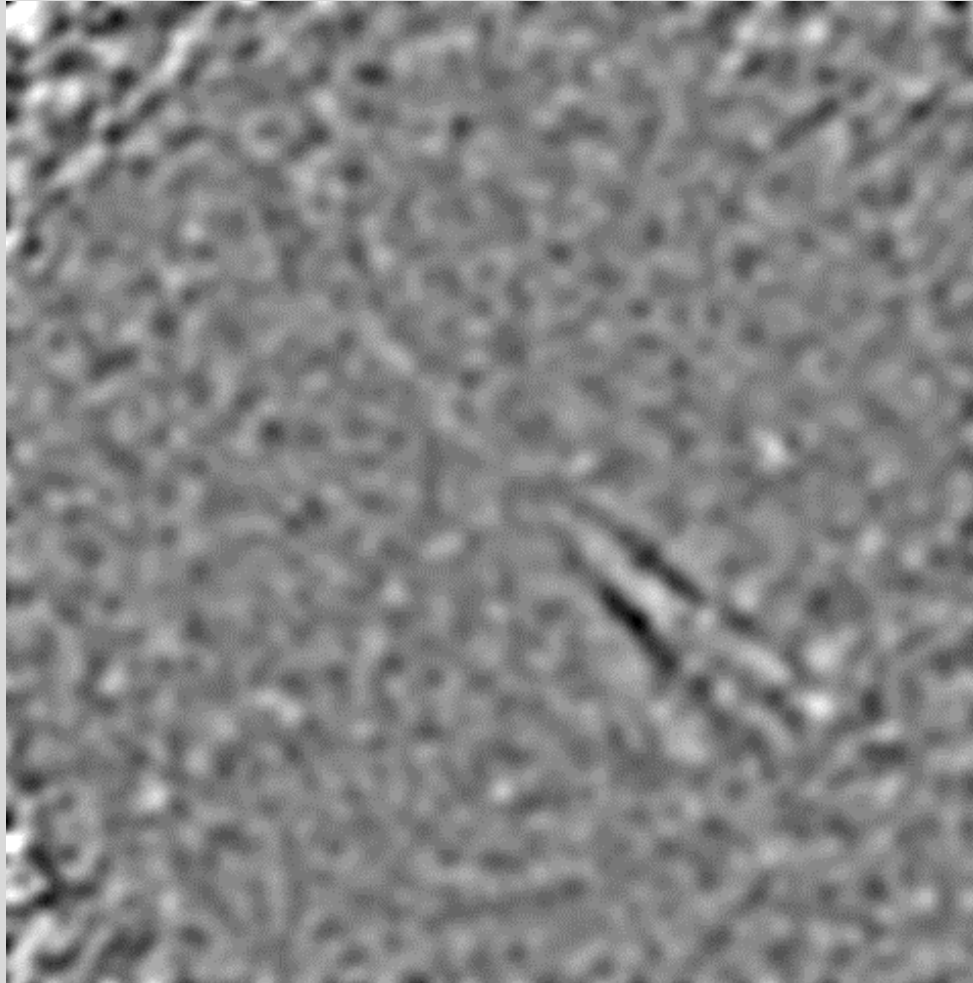
- Olfactory receptor neurons expressing the same receptor converge to the same glomerulus in the olfactory bulb



- Olfactory receptor neurons project to the olfactory bulb, forming a structure called a glomerulus.
- Each glomerulus receives input from a single class of olfactory receptor.
- An single odorant might active different types of receptors.

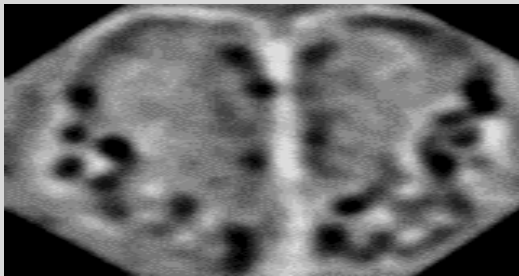


Glomerular activation in response to ethyl butyrate 0.1%

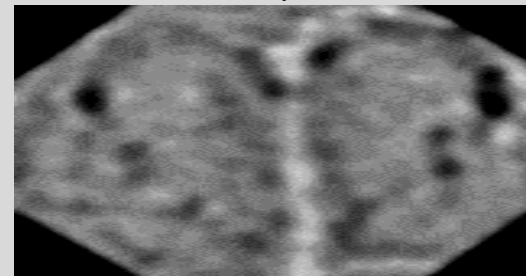


Odors activate overlapping patterns of olfactory receptors

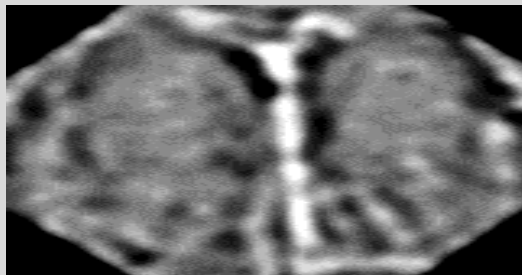
Ethyl
valerate



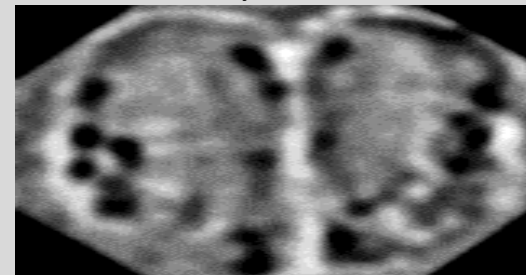
Ethyl
caproate



Isoamyl
acetate



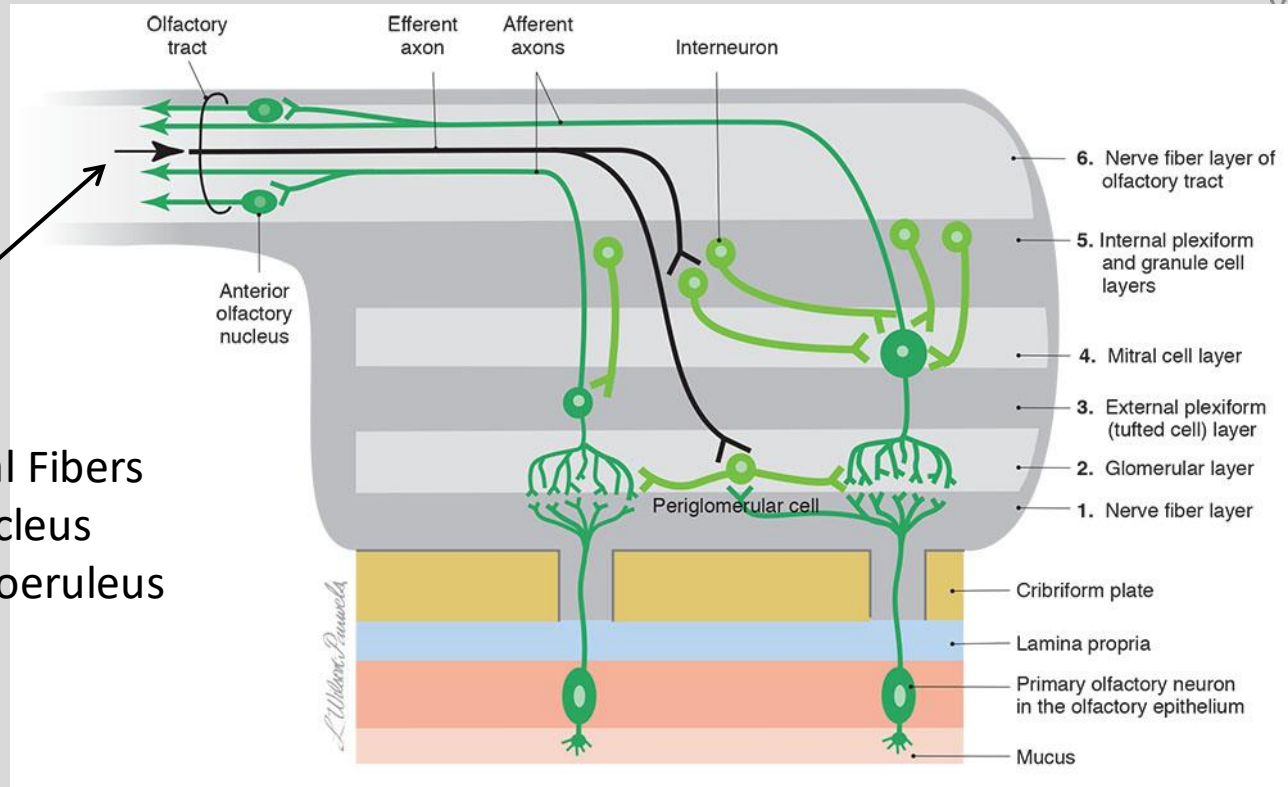
Ethyl
Butyrate



- Different odors can activate overlapping sets of glomerulus.
- The combination of olfactory glomeruli determines the identity of an odor.



- Olfactory bulb projections



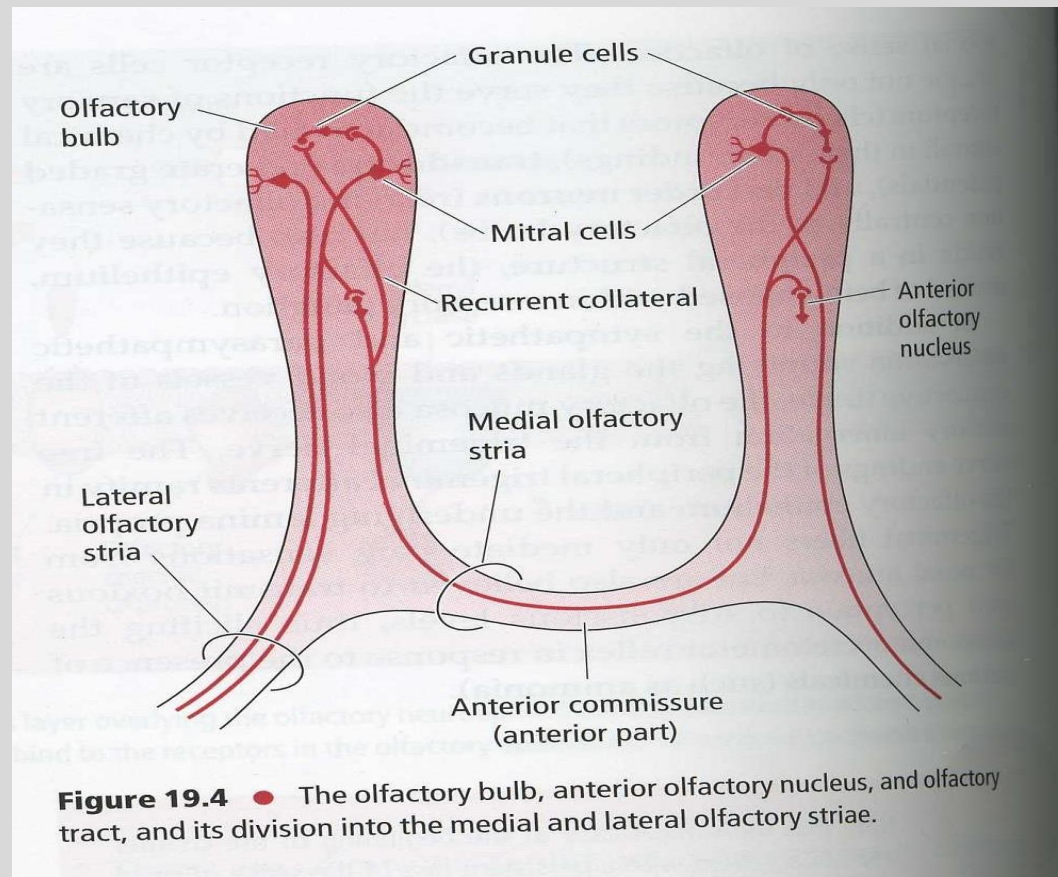
- Centrifugal Fibers
- Raphe Nucleus
- Nucleus Coeruleus

- Glomeruli are contacted by dendrites of mitral and tufted cells.
- These cells are the output of the bulb and transmit the olfactory information to other parts of the brain.
- The activity of mitral and tufted cells are regulated by interneurons of the olfactory bulb: granule cells and the periglomerular cells.
- These interneurons are regulated by descending projections from piriform cortex, anterior olfactory nucleus, and neuromodulatory systems.

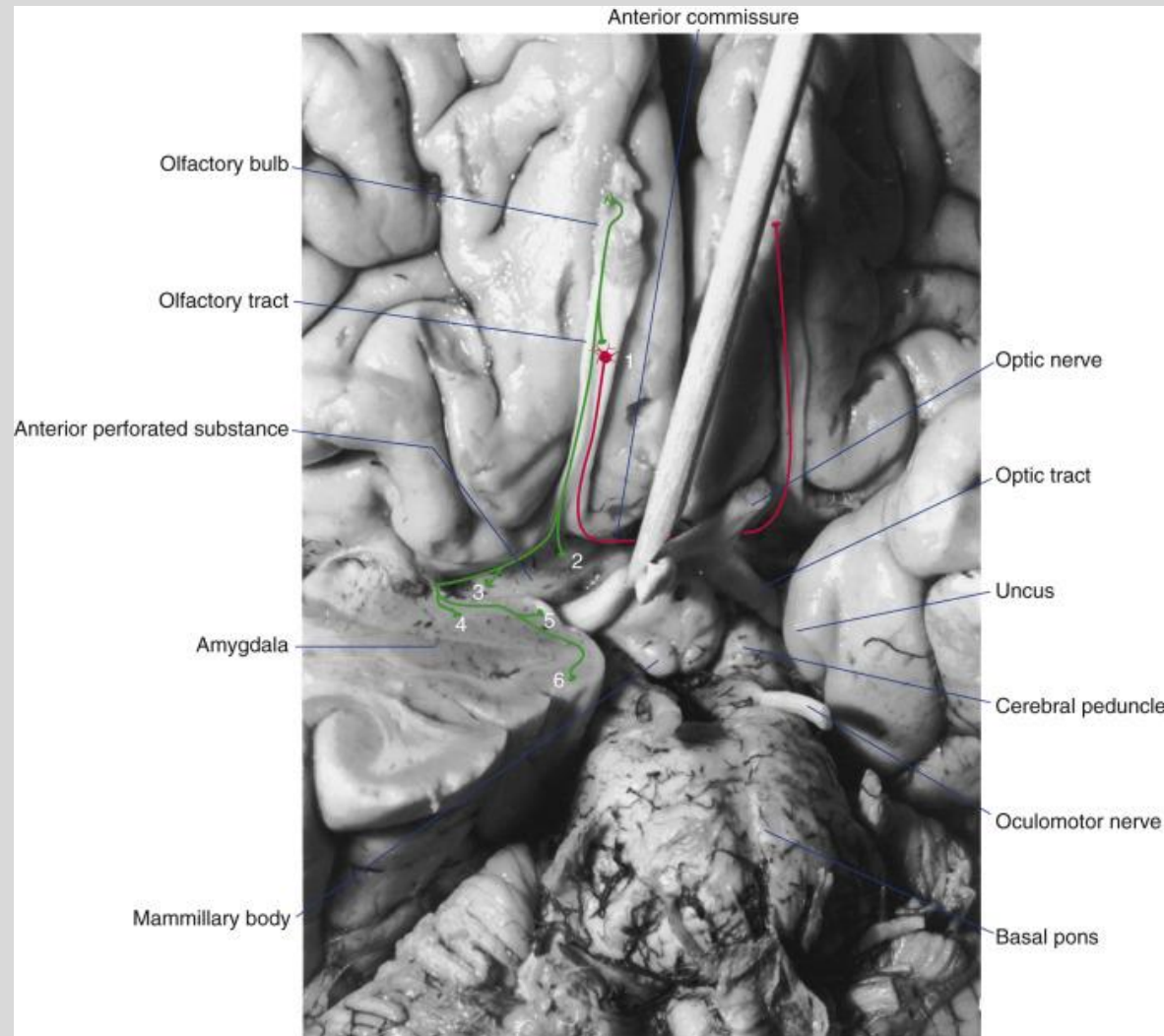


Olfactory Tracts

- Axons of mitral and tufted cells
- Olfactory tract expands at the Olfactory trigone
- Bifurcates into the Lateral and Medial stria
- Lateral tract has the axons of the principal cells and projects to the Primary olfactory cortex
- Medial olfactory stria project to the contralateral anterior olfactory nucleus.
- Anterior olfactory nucleus is an intermediate nucleus located in the olfactory tract.
- Anterior olfactory nucleus receives information from both bulbs and might play a role in olfactory localization.



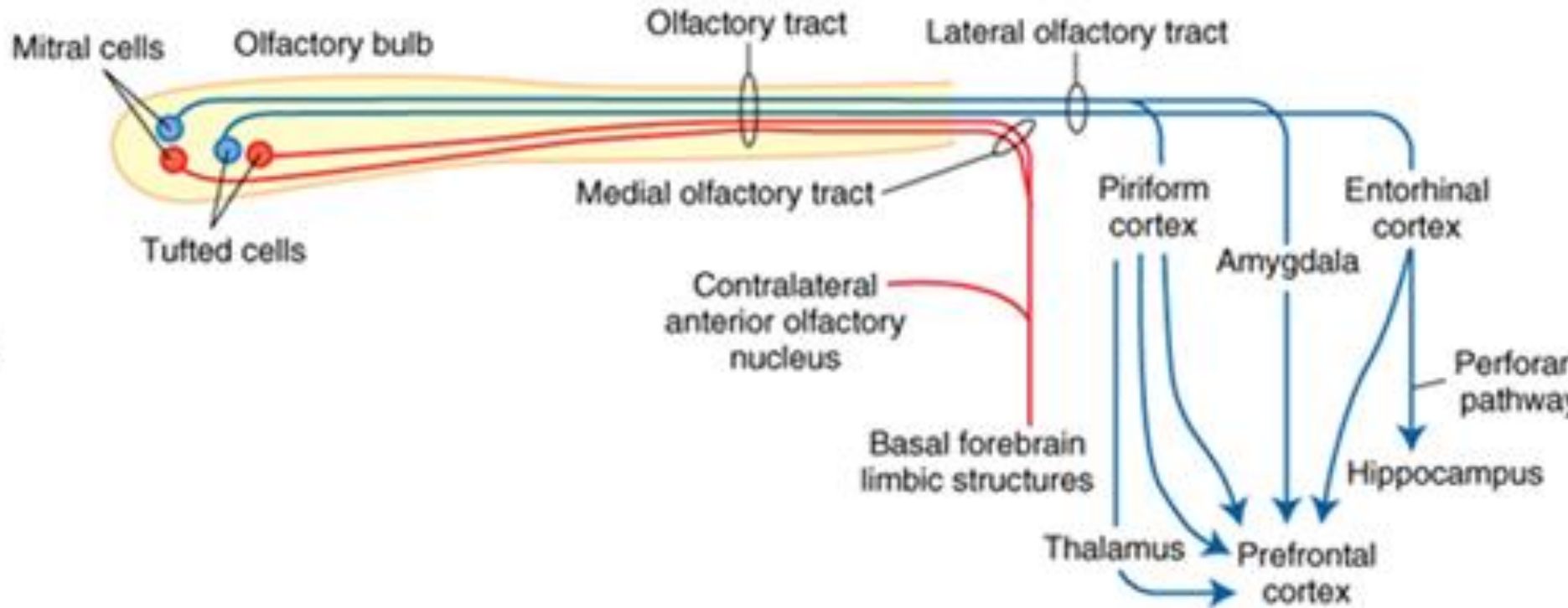
Olfactory Tracts



1. anterior olfactory nucleus
2. olfactory tubercle
3. piriform cortex
4. part of the amygdala
5. periamygdaloid cortex
6. parahippocampal gyrus, part of the entorhinal cortex



Olfactory bulb projections



- Anterior olfactory nucleus
 - Connects the two olfactory bulbs through a portion of the anterior commissure
- Piriform cortex (Part of the **Limbic System**)
 - Entorhinal area (Brodmann's area 28)
 - Dorsal entorhinal area (Brodmann's area 34)
 - Amygdala

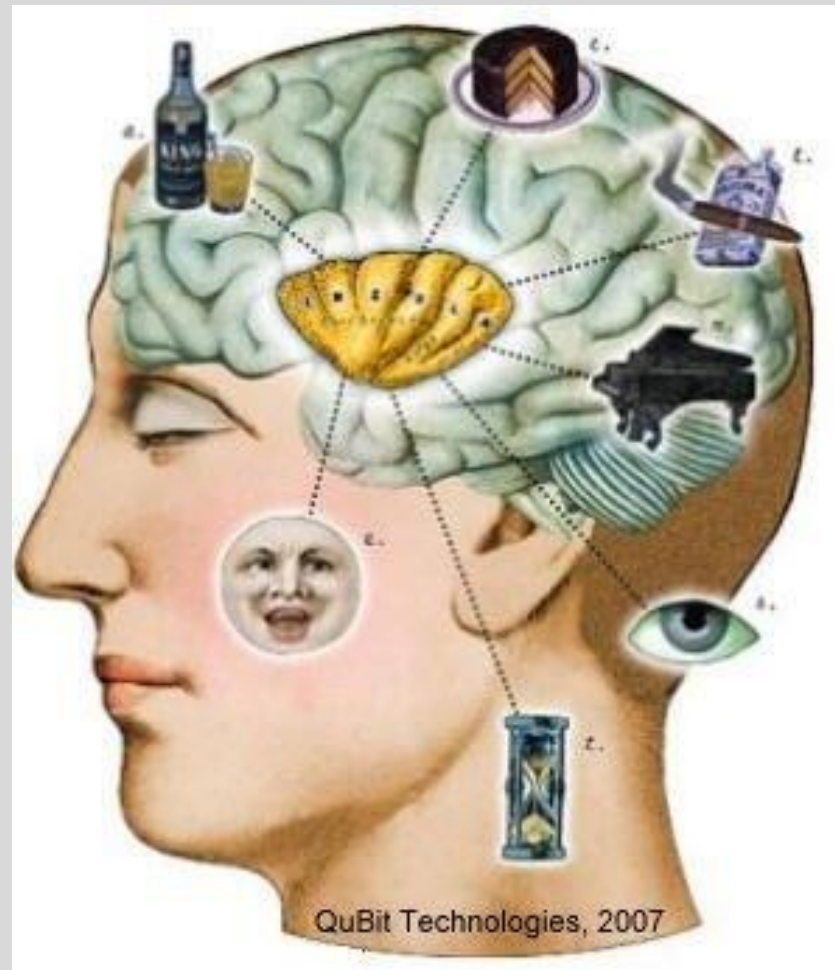


Mediodorsal Thalamic Nucleus

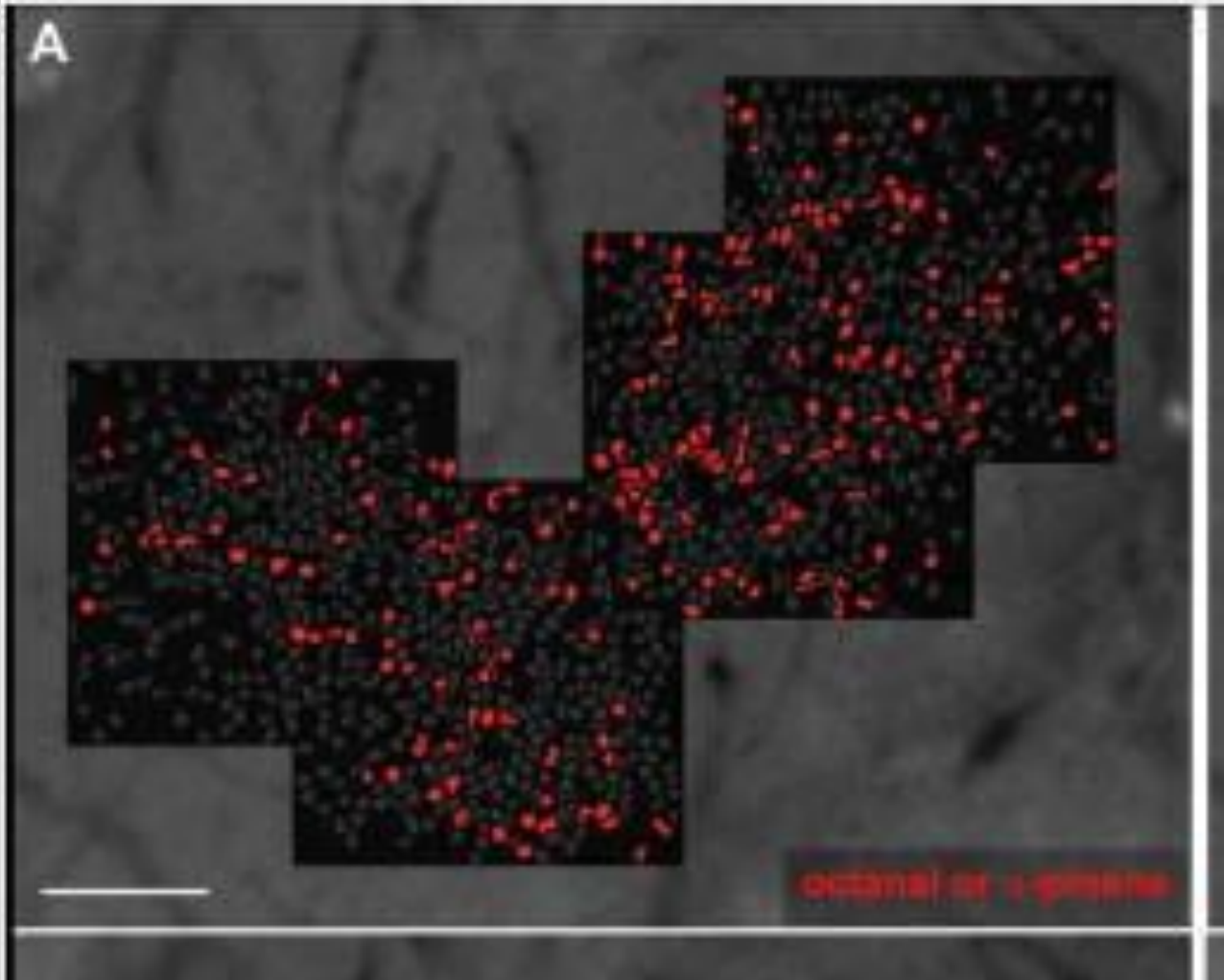
Receives projections from
piriform
cortex:"olfactory"
thalamus

Projects to:

- Insular cortex
- **Integration of smell & taste**
- Orbitofrontal cortex
- **Reward & motivation**



Glomerular representation of odors is lost in piriform cortex



Qualitative Olfactory Disorders

1) Parosmia: Stimulus elicits false sensation

- smelling burnt paper instead of baby powder
- smelling gasoline instead of coffee
- Typically, the vast majority of patients describes these parosmic experiences as unpleasant
- Parosmia is associated with loss of olfactory function
- Parosmia might present after COVID 19 induced anosmia(7.1% of patients)
- Cacosmia term is used when the perceived smell is unpleasant

2) Phantosmia: Random sensation of smell in the absence of stimulus

- Phantosmia is NOT associated with loss of olfactory function



Taste

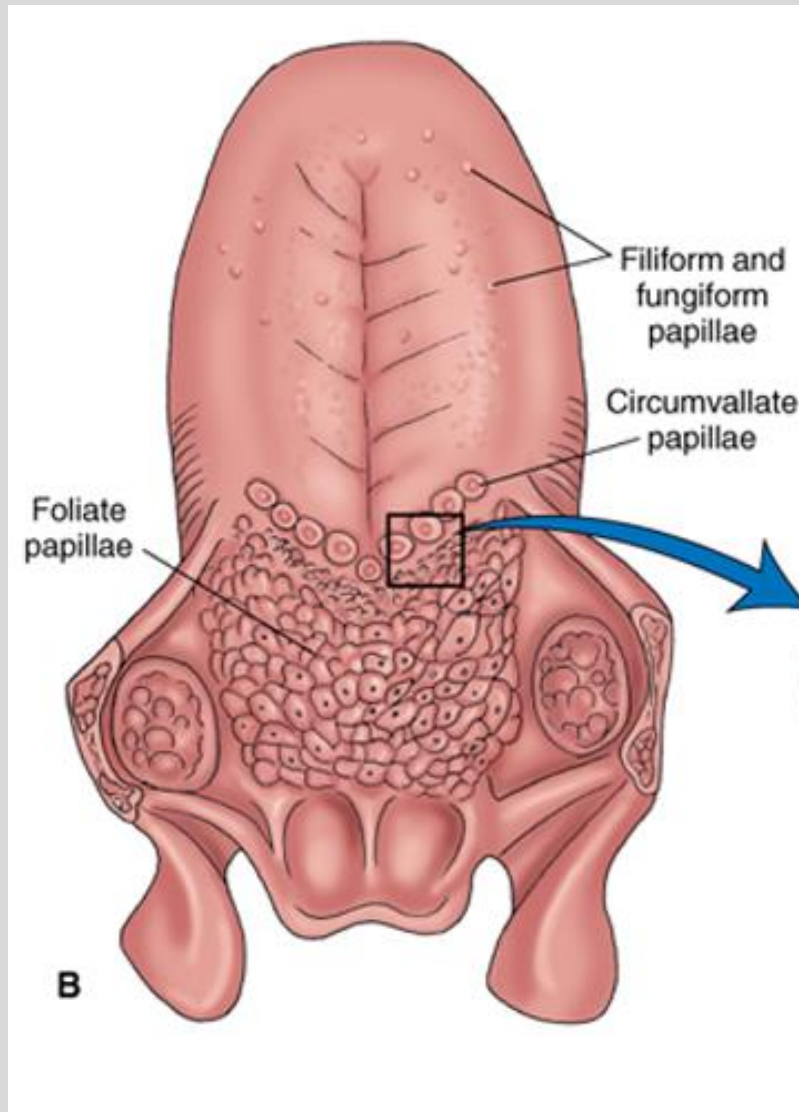
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Taste receptors in the tongue



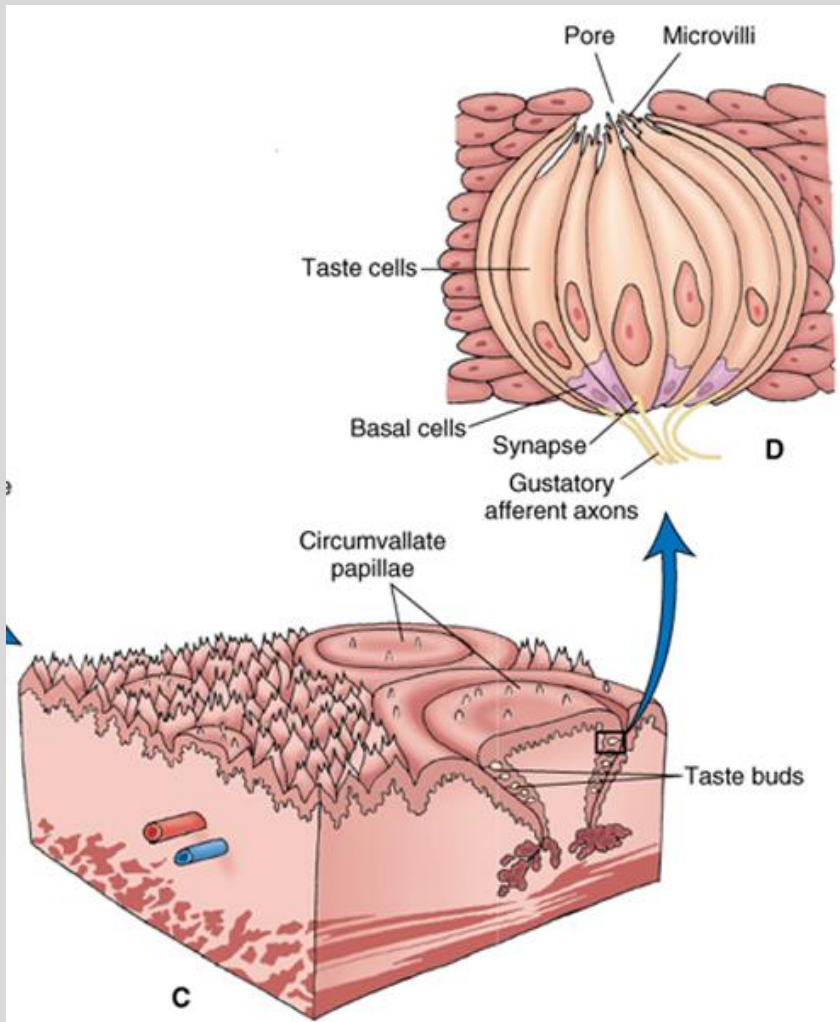
Five basic sensations of taste: sweet, bitter, salty, sour, and umami. Umami is a Japanese word for pleasant savory taste. Glutamates and nucleotides umami taste sensation.

The sensation of taste are located in taste buds, which are the sensory organs for the taste system.

Source: Siegel , Essential Neuroscience



Taste buds contain the gustatory receptors



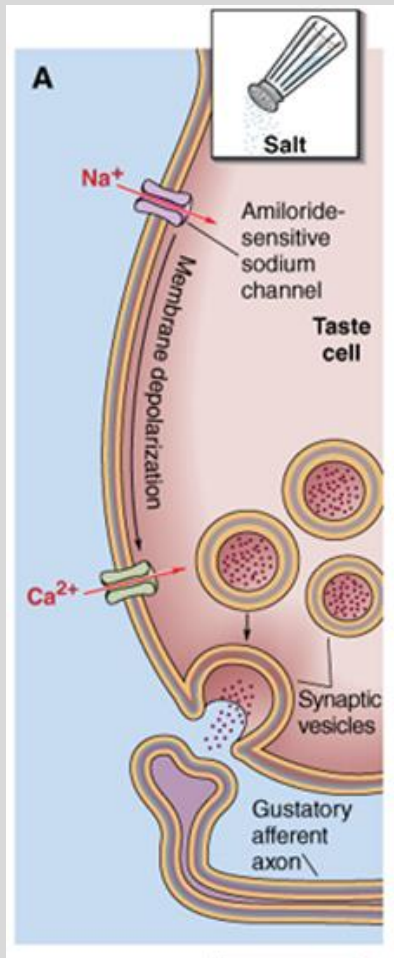
Taste buds are located in different types of papillae, which are protrusions on the surface of the tongue.

filiform, fungiform, foliate, and circumvallate papillae. The filiform and fungiform papillae are scattered throughout the surface of the anterior two thirds of the tongue, especially along the lateral margins and the tip.

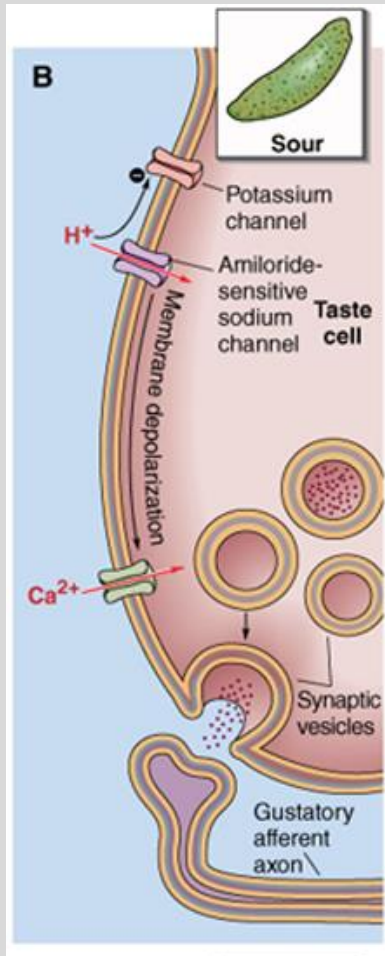


Different tastes have different mechanisms of sensory transduction

- The transduction of salt taste is mediated by the influx of Na^+ (sodium) through the amiloride-sensitive Na^+ channels.
- Depolarization of the membrane opens voltage-activated calcium channels.
- Calcium triggers release of synaptic vesicles, activating gustatory afferent axons.



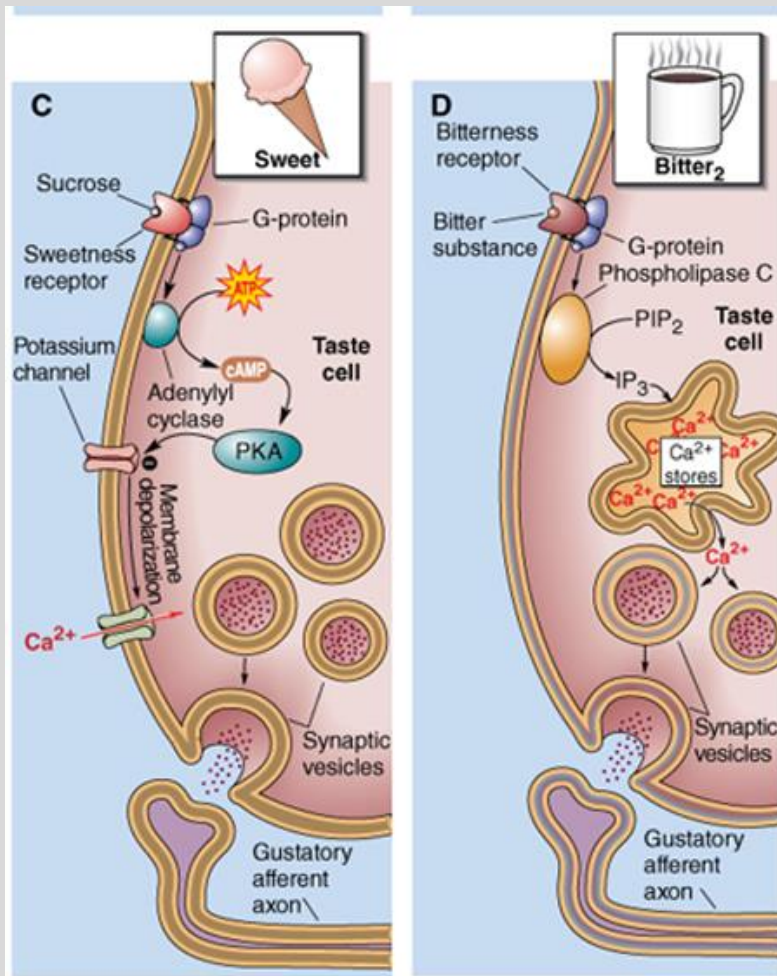
Different tastes have different mechanisms of sensory transduction



- Sour taste receptors detect protons H^+ , and is mediated by closure of voltage-dependent K^+ (potassium) channels.
- Depolarization of the membrane opens voltage-activated calcium channels.
- Calcium triggers release of synaptic vesicles, activating gustatory afferent axons.



Different tastes have different mechanisms of sensory transduction



-Sweet and bitter taste use G protein coupled receptors.

-Sweet taste is mediated by activation of G proteins and adenylate cyclase, which increases the levels of cyclic adenosine monophosphate (cAMP), producing membrane depolarization.

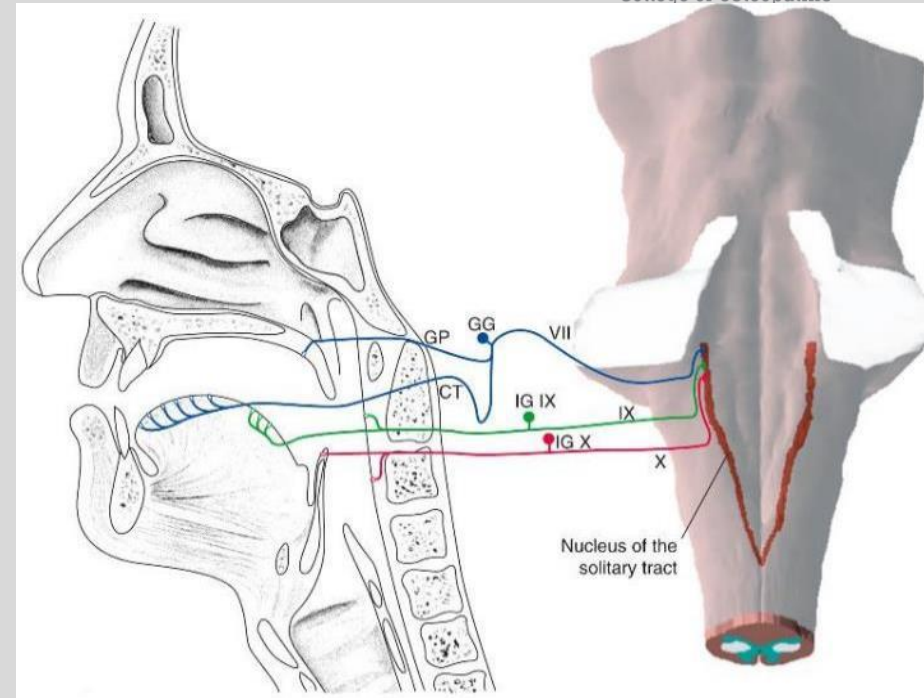
-Bitter substances activate a G protein that activates phospholipase C and generates inositol triphosphate (IP₃) and producing calcium release from internal stores.

-Very different chemicals can taste bitter and are potentially toxic. There are more than 30 different bitter receptors.

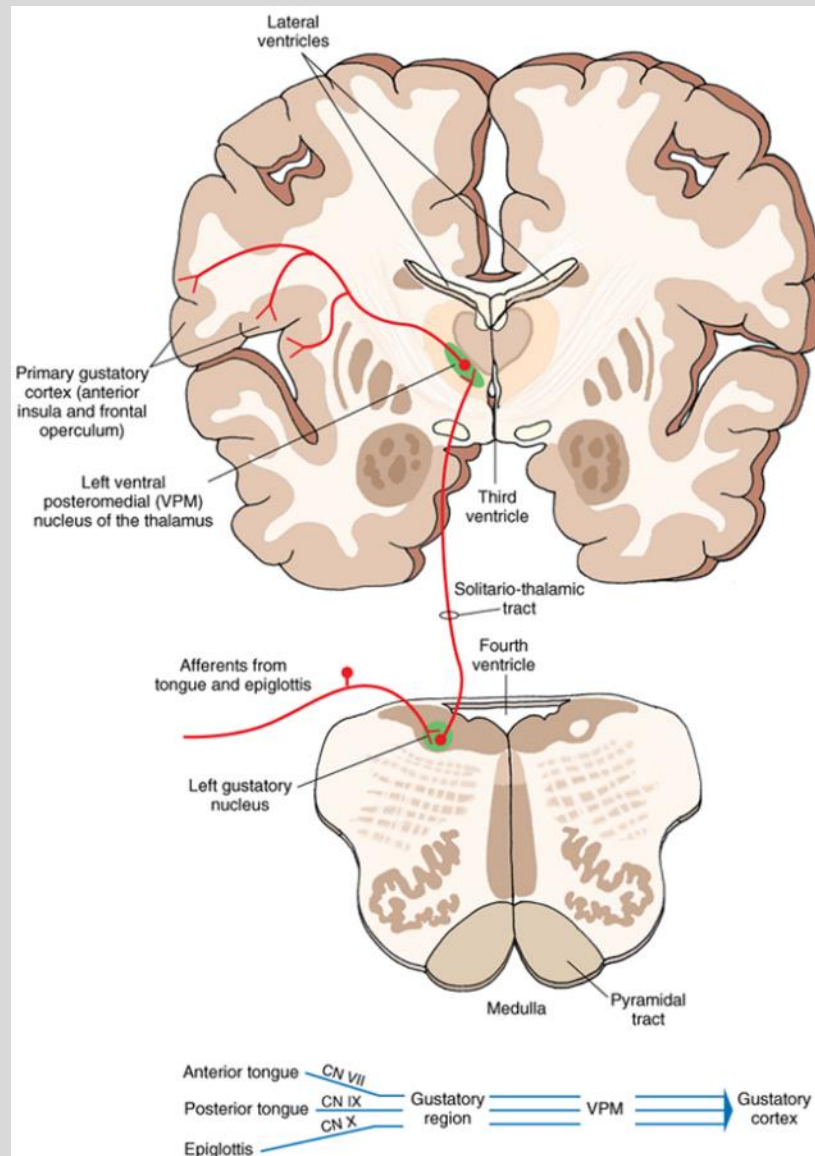


Taste is transmitted by CN VII, IX, X

- The taste buds on the anterior two thirds of the tongue are innervated by the facial nerve (CN VII).
- Taste buds on the posterior one third of the tongue are innervated by the glossopharyngeal nerve (CN IX).
- Taste buds on the epiglottis and pharyngeal walls are innervated by the vagus nerve (CN X)
- The central processes of all three terminate in rostral parts of the nucleus of the solitary tract in the dorsal medulla, close to the 4th ventricle.



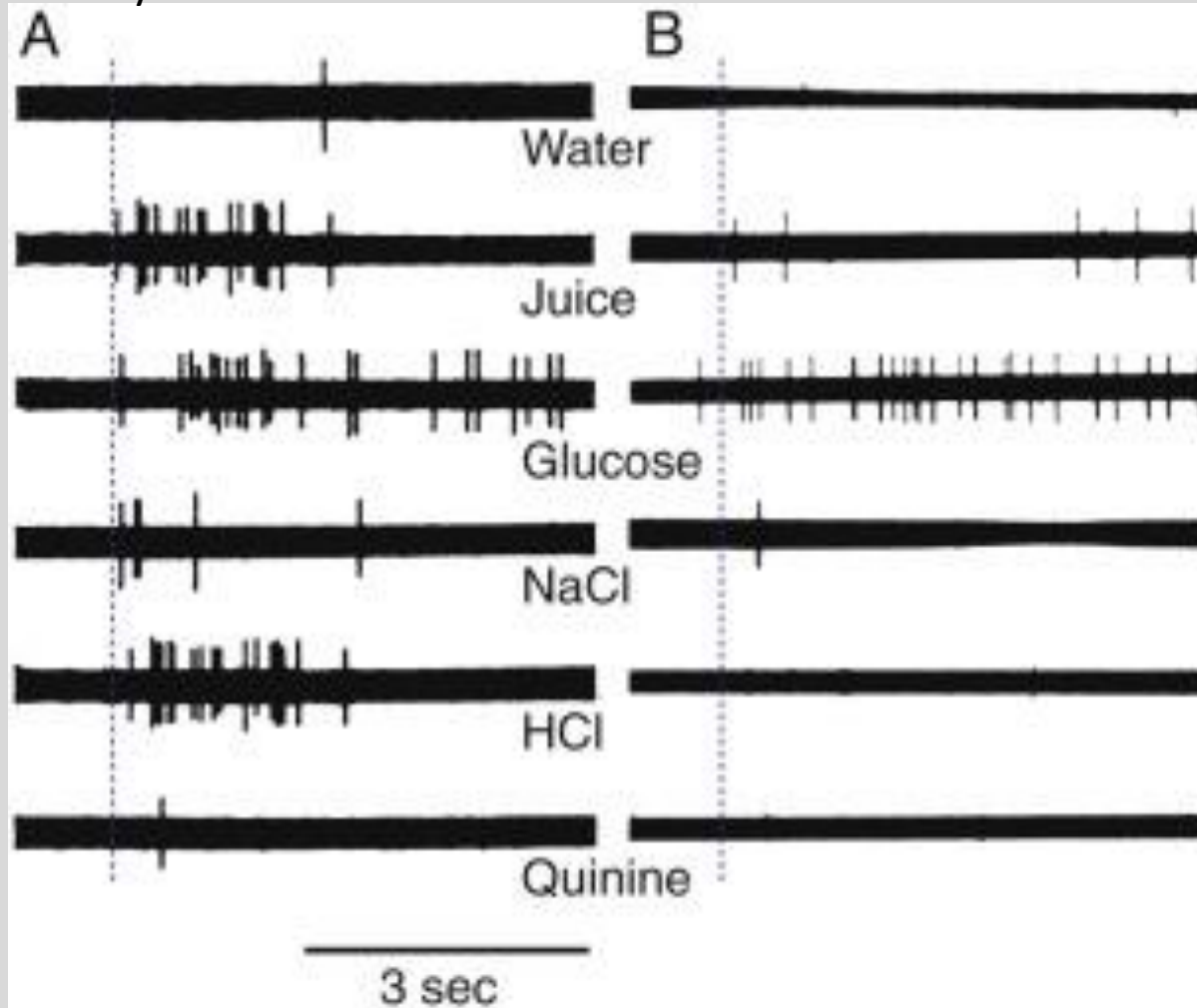
Taste Pathways



Solitary tract nucleus cells respond to multiple tastes compared to gustatory cortex

Solitary tract nucleus cells

Gustatory cortex



• Summary Slide

The olfactory nerve (CNI) consists of sensory neuron axons that synapse in the olfactory bulb. CNI carries only one single modality: Special Visceral Afferent.

There is no incoming precortical relay of olfactory information in the thalamus. Then, olfactory information travels via the mediodorsal nucleus of the thalamus to the insula followed by the orbitofrontal cortex.

Neurodegenerative diseases, such as Alzheimer's and Parkinson's disease, are often accompanied by hyposmia.

Each taste bud harbors all five varieties of non-neuronal sensory cells that are connected to either CNVII, IX or X.



